

Flatness of representation counts for prime sequences, I: the sub-peak spectrum and the second-order ripple

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Abstract

Let B be a finite set of odd primes and $r_B(m)$ the number of subsets of B summing to m . The companion paper reduces the asymptotic behaviour of the subset-sum landscape of B to a single quantity, the *flatness* ε_d , which measures the pointwise oscillation of r_B on a bounded window. This paper develops the Fourier-side input needed to bound ε_d : we determine the complete spectrum of sub-peaks of the characteristic product F_B , we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with m odd is exactly 0 whenever $m \in B$; for B a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \dots, 40$. Key steps are formally verified in Lean 4.

1 Introduction

1.1 The quantity to be bounded

For a finite set B of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \quad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta} d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant provided the *flatness*

$$\varepsilon_d = \max_{m, m' \in \text{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where B_d is the subsequence of elements exceeding $2d$ and Win_d is a bounded window about $T/2$. Bounding ε_d is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

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1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a \in B} \cos(a\theta/2)$, so that $|F_B| = 2^{|B|} |G_B|$, the natural per-element height at modulus q is

$$M(q) = \left(\prod_{\substack{1 \leq r \leq q \\ \gcd(r, q) = 1}} |\cos(\pi r/q)| \right)^{1/\varphi(q)}.$$

Proposition 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ (m odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction K_2 and phase drift ψ ; Section 5 treats the main arc and identifies the first correction to the central density, resolving an unexplained constant of the companion paper. Section 6 reports the numerical verification.

1.3 Relation to the literature

Three bodies of work are close enough to require comment.

Oscillations in partition counts. Murthy and Brack [MB26] give a semiclassical, periodic-orbit account of oscillations in partition counting functions; conceptually this is the same mechanism as ours, a few isolated saddle contributions producing a visible ripple. Their detailed oscillation analysis is for partitions into distinct *squares*, where the relevant orbits are indexed by Pythagorean triples, and their treatment of prime partitions (their §7) is asymptotic rather than oscillatory. The limit also differs: they let the integer being partitioned tend to infinity with the part set fixed, whereas we let the *number of parts* k tend to infinity.

Smoothness of partition functions. Bateman and Erdős [BE56] prove $p_A(n-1)/p_A(n) \rightarrow 1$ for a *fixed infinite* set A of parts with *repetitions allowed*. Our situation differs in both respects: our part set is a finite truncation that grows with k , and parts are used at most once, so the generating function is $\prod_{a \in B} (1 + x^a)$ rather than $\prod_{a \in A} (1 - x^a)^{-1}$. We do not see how to deduce the finite-truncation statement from theirs. The same remark applies to Roth and Szekeres [RS54] and to Vaughan [V08], both of which treat the fixed infinite set of all primes.

Landscapes of the number partitioning problem. Ferreira and Fontanari [FF98] compute the fraction of metastable states — that is, $\text{lm}/2^k$ — analytically for uniform random weights; Alyahya and Rowe [AR14] give a formula in terms of the coefficient of variation; Stadler, Hordijk and Fontanari [SHF03] analyse barrier trees. All of these average over *random* weights, whereas the companion paper and this one treat a *specific* sequence and an order-sensitive invariant.

Finally, the anti-concentration literature (Littlewood–Offord and its inverse forms) bounds the *maximum* of r_B ; what we need is a two-sided bound on the *ratio* at nearby points, which does not follow from a one-sided bound.

1.4 Formal verification

Statements marked **name** are machine-checked in Lean 4 against Mathlib; the repository builds with no **sorry** and no additional axioms. The canon currently consists of 95 theorems.

2 The sub-peak spectrum

The two statements of this section are classical: Proposition 2.1 is the resultant $\text{Res}(\Phi_q, x+1)$ written out, and Lemma 2.2 is a standard exercise. We record both with proofs because we use the precise constants, and because the interpretation of $M(q)$ as the per-element height of a sub-peak of a subset-sum generating function seems not to have been made before.

Proposition 2.1 (Cyclotomic evaluation; classical). *For every integer $q \geq 3$,*

$$M(q) = \frac{1}{2} |\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q -th cyclotomic polynomial.

Proof. Put $\zeta = e^{2\pi i/q}$. For $1 \leq r \leq q-1$ we have $|1 + \zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r,q)=1} (x - \zeta^r)$, evaluating at $x = -1$ gives $\prod_{\gcd(r,q)=1} |1 + \zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r,q)=1} |\cos(\pi r/q)| = 2^{-\varphi(q)} |\Phi_q(-1)|$. Take $\varphi(q)$ -th roots. $\square$

Lemma 2.2 (Classical values). *For $q \geq 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with p an odd prime and $j \geq 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \geq 2$; and $\Phi_q(-1) = 1$ otherwise.*

Proof. We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if n has at least two distinct prime factors; for odd $n \geq 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even m , $\Phi_{2m}(x) = \Phi_m(x^2)$. If q is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with m odd then $\Phi_q(-1) = \Phi_m(1)$, which is p when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with m even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when m is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. $\square$

Theorem 2.3 (Extremality of modulus six). *$\sup_{q \geq 3} M(q) = \sqrt{3}/2$, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \leq 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.*

Proof. By Proposition 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in j with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2} p^{1/(p^{j-1}(p-1))}$; the exponent decreases in j , so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2} p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \dots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2} 3^{1/6} \approx 0.6005$. Combining the three families gives the statement. $\square$

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

3 The vanishing family and the surviving spectrum

Lemma 3.1 (Exact vanishing). *Let $q = 2m$ with m odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

Proof. The factor of F_B indexed by m is $1 + e^{i\pi} = 0$. $\square$

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of m , so if B is a set of primes and $m > 1$ then B meets that class if and only if m itself lies in B . This gives a complete description.

Corollary 3.2 (Surviving spectrum). *Let B_d be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly, provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod{6}$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.
2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly, for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd a . This is the Fourier form of the modulus-4 obstruction of the companion paper.
3. $|G_{B_d}(2\pi/(2p))| = 0$ exactly, for every prime p with $2d < p \leq x$.
4. For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.
5. For $q = 2m$ with m odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2}3^{1/6} + o(1))^b$.
6. For q odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is 0 by parity.

Remark 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed d , only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes ≥ 5 ; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

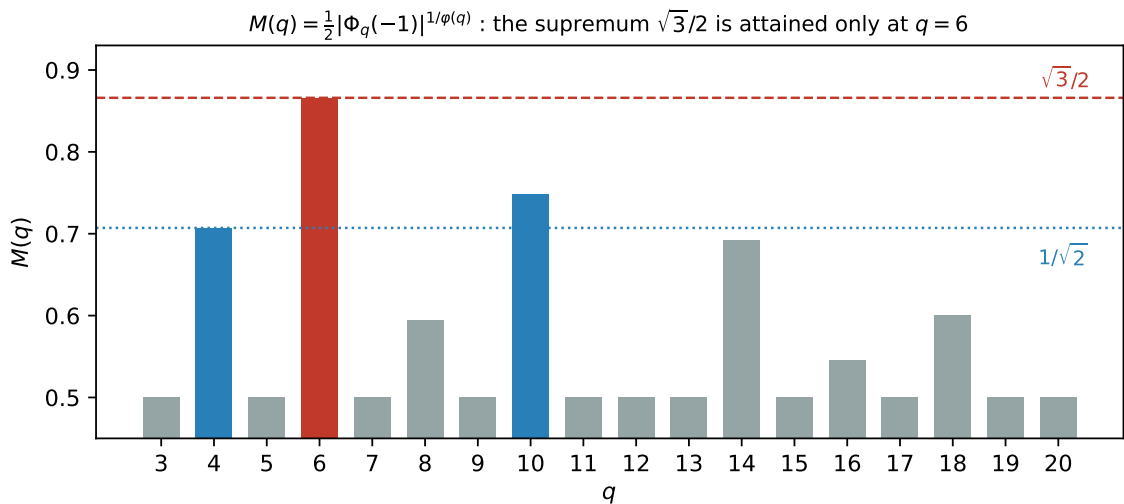


Figure 1: The per-element sub-peak height $M(q)$. The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

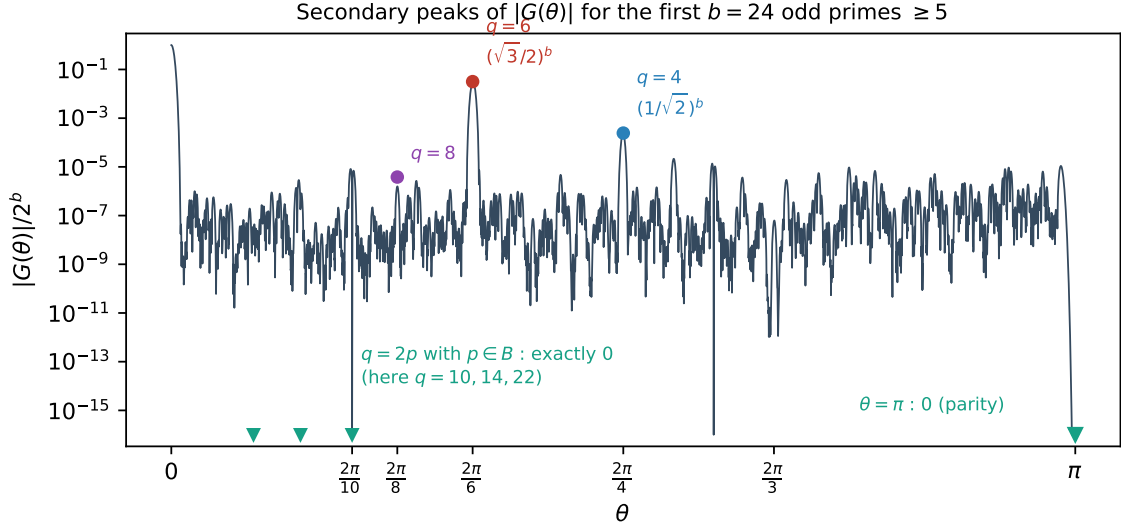


Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes ≥ 5 . The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1); $\theta = \pi$ vanishes by parity.

4 The modulus-six ripple to second order

Throughout this section B is a set of b odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a \in B} a^2, \quad S_4 = \sum_{a \in B} a^4, \quad \Delta = \sum_{a \in B} \tau_a a, \quad \Delta_3 = \sum_{a \in B} \tau_a a^3,$$

where $\tau_a = +1$ if $a \equiv 1 \pmod{6}$ and $\tau_a = -1$ if $a \equiv 5 \pmod{6}$, and let c_1, c_5 be the corresponding class counts. Write $\text{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

Lemma 4.1 (Local data at $\theta = \pi/3$). *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log |G_B(\pi/3 + t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

Moreover $\frac{d}{d\theta} \arg(1 + e^{ia\theta}) = a/2$ exactly, so the phase of F_B is linear along the arc and all curvature lives in $|G_B|$.

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_mul_div_six`.)

Proposition 4.2 (Second-order ripple formula). *For m in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\text{Main}(m)} - 1 = 2 \left(\frac{\sqrt{3}}{2} \right)^{b+1} K_2 \cos \left(\frac{\pi}{6} (2m - (c_1 - c_5)) + \psi(\nu) \right) + E,$$

where

$$K_2 = \exp \left[\frac{\Delta^2}{8S_2} - \frac{3S_4}{8S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^3} - \frac{3S_4\Delta^2}{16S_2^3} \right], \quad \psi(\nu) = -\frac{\sqrt{3}}{2} \nu \frac{\Delta + \Delta_3/S_2}{S_2}.$$

Derivation. Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L + i\nu)^2/(2V_6) + C_3(L + i\nu)^3/(6V_6^3) + \dots$, the Gaussian-fluctuation factor

$-\frac{1}{2}\log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \dots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives K_2 and ψ . The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b+1$. $\square$

Remark 4.3 (The two branches). The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \pmod{6}$; the term $-\frac{3}{8}S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \geq 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \leq 47$ and $K_2 < 1$.

Remark 4.4 (Error policy). The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in Δ , of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

5 The main arc and the central density

The main arc contributes the smooth part of r_B . The leading term is the local central limit theorem; what matters for the companion paper's absolute formula is the *first correction* to it.

Proposition 5.1 (Central density with first correction). *Let B be a set of b odd integers, $V_0 = S_2/4$, and let m be at the centre $T/2$. Then*

$$r_B(m) = \frac{2^b}{\sqrt{2\pi V_0}} \left(1 - \frac{S_4}{4S_2^2} + O((S_4/S_2^2)^2) \right).$$

Derivation. Write $\log G_B(\theta) = \sum_{a \in B} \log \cos(a\theta/2) = -\frac{V_0}{2}\theta^2 - \frac{S_4}{192}\theta^4 + O(S_6\theta^6)$, using $\log \cos u = -u^2/2 - u^4/12 + O(u^6)$ and $u = a\theta/2$. Since G_B is even (Lean: `G_even`), all odd-order terms vanish identically, so the first correction is the quartic one. Integrating $e^{-V_0\theta^2/2}(1 - \frac{S_4}{192}\theta^4)$ over the main arc and using $\int \theta^4 e^{-V_0\theta^2/2} = 3\sqrt{2\pi}V_0^{-5/2}$ gives the stated factor $1 - \frac{S_4}{192} \cdot 3V_0^{-2} = 1 - \frac{S_4}{4S_2^2}$. $\square$

Remark 5.2 (The absolute formula of the companion paper). The companion paper predicts the number of strict local minima by $\text{lm} \approx \Gamma \cdot 2^b/\sqrt{2\pi V_0}$, i.e. by replacing the ground-state count by its Gaussian estimate. The measured ratio of the two sides sits systematically below 1 — about 0.97 — with no explanation offered there. Proposition 5.1 identifies the deficit as $S_4/(4S_2^2)$, which at $k = 20$ equals 0.0246. Section 6.4 verifies this. We warn against one confusion: the value 0.972 appearing here (the ratio lm/pred at $k = 20$) is *not* the 0.972 in the companion paper's list of $Q(k)$, which is $Q(18)$. Those are different quantities and the numerical coincidence is accidental: Q is formed with the *measured* ground-state count, so it carries no Edgeworth correction, and it oscillates about 1 rather than sitting below it.

6 Numerical verification

All computations below are exact integer dynamic programming for r_B , followed by a discrete Fourier extraction; logs are included in the repository.

6.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

Δ_3^2 coefficient	$S_4\Delta^2$ term	branch A mean	branch B mean
45/24	no	0.00061	0.00736
45/24	yes	0.00060	0.00400
5/24	no	0.00073	0.00411
5/24	yes	0.00072	0.00073

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \dots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

On branch B the individual values of $\text{measured}/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$. The omitted contributions are dominated by the class-independent tail $O((S_4/S_2^2)^2)$, and S_4/S_2^2 decreases monotonically over the range, from 0.1111 at $k = 18$ to 0.0507 at $k = 40$; hence the residual is largest at the smallest k . That the Δ -dependent quantity Δ_3^2/S_2^3 is in fact largest at $k = 32$ ($2.70 \cdot 10^{-3}$, against $1.04 \cdot 10^{-3}$ at $k = 18$) without producing an excess residual there is corroborating evidence that the Δ -dependence is already absorbed into K_2 .

6.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \dots, 40$, only α is significant, at every window width tested: $t(\alpha)$ ranges from +4.26 to +5.16, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model $\text{measured} = c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths 24, 48, 96 and 192, with R^2 from 0.978 to 0.986; restricting to $|\Delta| \geq 17$ raises R^2 to 0.993. We conclude that ψ as stated is correct, with the deficit in c a finite-window effect.

Remark 6.1 (Extraction). The amplitude is extracted from a narrow window of fixed half-width 24, i.e. eight full periods of the modulus-6 ripple, which over $k = 18, \dots, 40$ is between 0.08 and 0.30 standard deviations; both requirements — enough periods, and narrower than σ — hold throughout. Wide windows corrupt the Gaussian reference. As a check, at $k = 18$ the ratio $\text{measured}/K_2$ is 0.9988, 0.9974, 1.0034 and 1.0065 at half-widths 12, 16, 24 and 32, so the window choice moves the residual by less than 1%. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

6.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level d : a peak at $q = 2p$ is dark exactly when p is still present in B_d , and lights up as soon as the truncation removes p . All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every d , confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and 2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter. This is to be expected, since the error in the prime number

d	removed primes	$q = 6$	$q = 10$	$q = 14$	tallest
1	—	0	0	0	$q = 4$
2	3	$3.66 \cdot 10^{-2}$	0	0	$q = 6$
3	3, 5	$4.22 \cdot 10^{-2}$	$1.03 \cdot 10^{-3}$	0	$q = 6$
4	3, 5, 7	$4.88 \cdot 10^{-2}$	$1.75 \cdot 10^{-3}$	$4.33 \cdot 10^{-4}$	$q = 6$

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for A the first 24 odd primes. Entries equal to 0 are exact zeros.

theorem in arithmetic progressions decreases only logarithmically, so at $b = 21, \dots, 31$ the $o(1)$ in items (iv)–(v) of Corollary 3.2 has not yet become small.

6.4 The central density

We test Proposition 5.1 using only ground-state counts. Write $\lambda(B) = \deg(B)\sqrt{2\pi V_0}/2^b - 1$ for the measured relative deficit, where $\deg(B)$ is the number of subsets attaining the minimum of $|\sum S - \lfloor T/2 \rfloor|$. Proposition 5.1 predicts $\lambda \approx -S_4/(4S_2^2)$.

k	deg	measured λ	predicted	difference
16	394	−0.024764	−0.031338	+0.006573
18	1290	−0.025573	−0.027754	+0.002181
20	4344	−0.024036	−0.024629	+0.000594
24	51068	−0.021467	−0.020830	−0.000637
28	634568	−0.017663	−0.017574	−0.000090

Table 3: Central-density deficit for the first k odd primes. The agreement is within 0.5% for $k \geq 18$ and improves to 0.01% at $k = 28$.

Over an ensemble of 100 random odd sequences per size (drawn from the same range as the primes), the mean absolute difference is 0.0171 at $k = 16$, 0.0079 at $k = 18$ and 0.0038 at $k = 20$; the ensemble is noisier than the primes because $\deg$ is a small integer subject to arithmetic fluctuation.

The consequence for Remark 5.2 is direct. Writing $\rho = \ln/(\Gamma \cdot 2^b/\sqrt{2\pi V_0})$ for the ratio in the absolute formula, the ensemble mean of ρ is 0.9462, 0.9621, 0.9672 and 0.9716 at $k = 12, 16, 18, 20$, while the mean of $\rho/(1 - S_4/(4S_2^2))$ is 0.9812, 0.9888, 0.9912 and 0.9933. The correction removes about three quarters of the deficit, and the share it removes grows with k . We therefore record the systematic part of the 0.97 as explained; the residual 0.7% at $k = 20$ is a separate and smaller effect, still decreasing.

7 A free Gaussian envelope at every peak

The following is elementary but useful: every sub-peak carries an envelope as sharp as the main arc, with no arithmetic input at all.

Proposition 7.1 (Curvature bound). *Let B be any finite set of positive reals and $V_0 = \frac{1}{4} \sum_{a \in B} a^2$. On any interval where G_B has no zero,*

$$\frac{d^2}{d\theta^2} \log |G_B(\theta)| = - \sum_{a \in B} \frac{a^2}{4} \sec^2\left(\frac{a\theta}{2}\right) \leq -V_0,$$

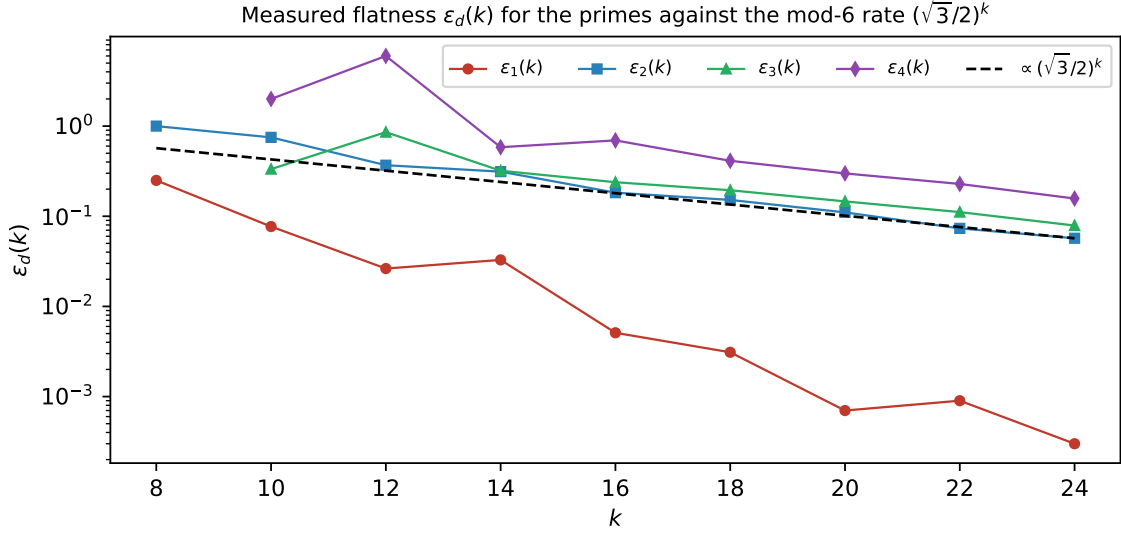


Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1).

because $\sec^2 \geq 1$. Consequently $\log |G_B|$ is concave there with curvature at least V_0 , and if θ^* is an interior local maximum then

$$|G_B(\theta)| \leq |G_B(\theta^*)| e^{-V_0(\theta-\theta^*)^2/2}$$

throughout that interval.

(Lean: `one_le_one_div_cos_sq`, `term_curvature_ge` and `curvature_ge_V0` formalise the inequality $V_0 \leq \sum(a^2/4) \sec^2(a\theta/2)$; the passage from concavity to the quadratic bound is standard and not yet formalised.)

Two caveats matter in practice, and both are visible numerically.

Remark 7.2 (The centre is not $2\pi/q$). The envelope must be anchored at the true local maximum, which is displaced from $2\pi/q$ by $-h'(0)/h''(0)$ with $h = \log |G_B|$; by Lemma 4.1 this is $\asymp \Delta/S_2$ at $q = 6$. For the first 24 odd primes truncated at $d = 2$ the measured displacements are $+1.316 \cdot 10^{-5}$ ($q = 6$), $+3.795 \cdot 10^{-4}$ ($q = 4$) and $-2.097 \cdot 10^{-3}$ ($q = 8$), against predicted $+1.316 \cdot 10^{-5}$, $+3.800 \cdot 10^{-4}$ and $-2.021 \cdot 10^{-3}$. Anchoring at $2\pi/q$ instead makes the envelope fail, by $7.6 \cdot 10^{-6}$ at $q = 6$ and $4.7 \cdot 10^{-3}$ at $q = 4$ in the same data.

Remark 7.3 (Validity radius). The bound holds only inside the zero-free interval, whose half-width is $\approx \pi/(2 \max B)$. Measured in units of the envelope scale $V_0^{-1/2} = 2/\sqrt{S_2}$, this is

$$\frac{\pi}{2 \max B} \cdot \frac{\sqrt{S_2}}{2} \approx \frac{\pi}{4} \sqrt{k/3} \approx 0.45\sqrt{k},$$

so the envelope covers a growing number of standard deviations: 1.72 at $k = 16$, 2.08 at $k = 24$, 2.71 at $k = 40$ and 2.92 at $k = 48$, against the prediction 1.80, 2.21, 2.85, 3.12.

8 Effective constants for the surviving peaks

Items (iv) and (v) of Corollary 3.2 are the only ones that use equidistribution, and only for a finite list of moduli. We make them effective.

Write $n_r = \#\{p \in B_d : p \equiv r \pmod{q}\}$ for $\gcd(r, q) = 1$, and $e_r = n_r - b/\varphi(q)$. Since $\log |G_{B_d}(2\pi j/q)| = \sum_r n_r \log |\cos(\pi r j/q)|$ and $\sum_r e_r = O(1)$,

$$|G_{B_d}(2\pi j/q)| \leq M(q)^b \exp\left(L_q \sum_r |e_r|\right), \quad L_q = \max_{\gcd(r, q)=1} |\log |\cos(\pi r/q)||.$$

Bennett, Martin, O'Bryant and Rechnitzer [BMOR18] supply two explicit inputs which between them cover all x : their Theorem 1.3 gives $|\pi(x; q, a) - \text{Li}(x)/\varphi(q)| < c_\pi(q) x/(\log x)^2$ for $x \geq x_\pi(q)$, with $c_\pi(q) \leq 1/840$ and $x_\pi(q) \leq 8 \cdot 10^9$ for $3 \leq q \leq 10^4$; and their Theorem 1.9 gives $\max_{y \leq x} |\pi(y; q, a) - \text{Li}(y)/\varphi(q)| \leq 2.734\sqrt{x}/\log x$ for $x \leq x_2(q)$, where $x_2(q) = 10^{13}$ for $5 < q \leq 100$. The ranges overlap, so there is no gap.

Proposition 8.1 (Effective form of Corollary 3.2(iv)–(v)). *For each $\eta > 0$ and each modulus q in the finite list there is an explicit $k_0(q, \eta)$ such that $|G_{B_d}(2\pi j/q)| \leq (M(q) + \eta)^b$ for all $k \geq k_0(q, \eta)$. With the constants of [BMOR18] one may take*

q	4	6	8	10	14	18
$k_0(q, 0.10)$	108	34	1694	856	3547	4411
$k_0(q, 0.05)$	343	104	5596	2839	11914	14723
$k_0(q, 0.01)$	5984	1737	103960	52658	226200	278350

Remark 8.2. The $\sqrt{x}$ -shaped bound of [BMOR18, Thm. 1.9] is what makes these thresholds small. Using only [BMOR18, Thm. 1.3] one is forced to $x \geq x_\pi(q) \approx 10^9$, i.e. $k_0 \approx 4 \cdot 10^8$; the two theorems must be combined. On the other hand, even $k_0 \approx 10^2$ is far beyond the range $k \leq 44$ of the computations in Section 6, so Proposition 8.1 is genuinely asymptotic and does not certify the measured values. We state it as an effective bound, not as an explanation of the data. Note also that $q = 4$ and $q = 6$ appear in the table only for comparison: for those two moduli Corollary 3.2(i)–(ii) are exact identities and no equidistribution input is needed at all.

9 What remains

The results above supply the arithmetic input for a circle-method treatment of ε_d . Three steps remain, and we state them explicitly.

Problem 9.1 (Main arc). Make the local central limit theorem for r_B on the main arc effective, with an error term of the shape C/S_2 arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in b .

Problem 9.2 (Minor arcs). Bound the contribution of θ away from the surviving peaks. By Corollary 3.2 and Proposition 8.1 it suffices to treat $q = 4$, $q = 6$, and a finite list depending on d . For the exponential sums that arise, the closest available technology is that of Dong, Robles, Zaharescu and Zeindler [DRZZ25], who handle the non-principal major arcs for prime partitions by way of what they call strange and pseudo-differentiable functions.

Problem 9.3 (Assembly). Combine the two to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \rightarrow 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for ε_d can be extracted from the aggregate residue-class spread. The route to Problem 9.3 must go through the upper bounds above.

10 Appendix: the companion paper's Problem on random comparison

The companion paper asks whether $\ln P_k / \ln R_k \rightarrow 0$, where P_k is the first k odd primes and R_k a random odd sequence of the same length drawn from the same range, and reduces this to comparing $\Gamma(P_k)$, which converges, with $\mathbb{E}[\Gamma(R_k)]$. The expectation is exactly computable.

Theorem 10.1. *Let R_k be a uniformly random k -subset of the N odd integers in $[3, p_k]$, where $N = (p_k - 3)/2 + 1$, and put $c = (N + 1)/(k + 1)$. Then*

$$\mathbb{E}[\Gamma(R_k)] = (1 - 2^{-k}) + 2c(2 - (k + 2)2^{-k}).$$

In particular $\mathbb{E}[\Gamma(R_k)] = 1 + 4c + O(k2^{-k})$, and since $c \sim p_k/(2k)$, this diverges as soon as $p_k/k \rightarrow \infty$.

Proof. Identify the odd integers in $[3, p_k]$ with $\{1, \dots, N\}$ via $a = 2i + 1$. For a uniform k -subset the order statistics satisfy $\mathbb{E}[X_{(j)}] = j(N + 1)/(k + 1)$ exactly, so $\mathbb{E}[a_{(j)}] = 1 + 2jc$. Hence $\mathbb{E}[\Gamma] = \sum_{j=1}^k (1 + 2jc)2^{-j} = (1 - 2^{-k}) + 2c(2 - (k + 2)2^{-k})$, using $\sum_{j \leq k} j2^{-j} = 2 - (k + 2)2^{-k}$. $\square$

Remark 10.2. No prime number theorem is needed: only $p_k/k \rightarrow \infty$, i.e. that the primes have density zero, which is elementary. Monte-Carlo agreement with Theorem 10.1 is within one standard error at $k = 100, 400, 800$ ($z = -1.46, -0.38, +1.07$ over 20,000 samples).

For a statement in probability rather than in mean one needs a lower tail bound, and the following suffices.

Proposition 10.3 (Sketch). *Fix J and $\delta > 0$. Then $\mathbb{P}[a_{(J)} \leq \delta N/k] \leq (e\delta/2J)^J$, whence $\Gamma(R_k) \geq 2^{-J}\delta N/k$ with probability at least $1 - (e\delta/2J)^J$. Choosing $\delta = 1/2$ and then J large gives $\Gamma(R_k) \geq c_0 \log k$ with probability $1 - o(1)$, and therefore $\Gamma(P_k)/\Gamma(R_k) \rightarrow 0$ in probability.*

The tail bound was checked over 48 combinations $k \in \{100, 400, 1600, 3200\}$, $J \in \{1, 2, 4, 8\}$, $\delta \in \{0.25, 0.5, 1\}$ with 10^4 samples each; it held in every case, with the measured probability between one half and one seventh of the bound. A complete proof will appear with the assembly of Problem 9.3.

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